

1 Product Overview

The portable integrated UAV detection and countermeasure device integrates detection, tracking, positioning, countermeasure and networking functions. A single device can complete the full UAV control workflow. With sufficient endurance, lightweight construction and reinforced design, it is well suited for backpack or handheld use.

The device scans UAV spectrum characteristics and decodes protocol messages to detect unauthorized UAV intrusions. It can identify control and video-transmission signals between UAVs and controllers, and uses electromagnetic suppression to interfere with UAV remote-control, video-transmission and navigation signals, forcing UAVs to return or land.



Figure 1 Portable Integrated UAV Detection and Countermeasure Device

2 Key Features

- Integrated Detection and Countermeasure

Integrated detection and countermeasure design with high integration; one device can complete UAV detection and response.

- Broad UAV Model Detection

Supports detection and identification of consumer, industrial and non-standard UAVs in complex electromagnetic environments, with low false-alarm and missed-alarm rates.

- Long Countermeasure Distance

Up to 3 km in open environments (typical target: Mavic 3).

- Directional Search

Uses passive RF detection technology to provide 360° omnidirectional detection while stationary; after warning, it can switch to directional detection to determine the UAV direction.

- Accurate Identification

Accurately identifies the body serial number, model, position, speed, altitude, flight track and controller position of mainstream UAV brands.

- Multiple Countermeasure Effects

Provides multi-band RF jamming capability to make UAVs return, land or hover.

- Portable Operation

Small, lightweight and easy to carry.

- Long Endurance

Can be equipped with a high-capacity rechargeable battery for long endurance.

Basic Specifications	
Detection Mode	Supports both omnidirectional detection and directional direction finding
Detection Frequency Band	433M/800M/900M/ 1.4G/2.4G/5.2G/5.8G
Direction-Finding Accuracy	$\leq 30^\circ$ (directional detection); $\leq 0.5^\circ$ (protocol decoding)
Maximum Detection / Direction-Finding Distance	≥ 3 km (in open environment)
Countermeasure Frequency Band	433M/800M/900M/ 1.2G/ 1.4G/ 1.6G/2.4G/5.2G/5.8G
Maximum Effective Jamming Distance	≥ 3 km (line-of-sight conditions)
Jamming-to-Control Range Ratio	$\geq 20:1$ (typical value: MAVIC 2)
Operating Time	Detection standby time ≥ 4 h; continuous countermeasure operation ≥ 30 min
Power Supply Method	Built-in lithium battery / external adapter
Operating Temperature	$-20^\circ\text{C} \sim +55^\circ\text{C}$
Product Form	Shield-shaped design
Networking Mode	Can connect to a unified UAV platform
Operating Humidity	$\leq 95\%$
Control Mode	10.1-inch touchscreen operation

Maximum Total Power Consumption	500W
Protection Rating	IP65
Weight	8KG